

Tristan Misko

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SUMMARY

Sole engineer on a Berkeley research platform that publishes a policy index: Flask and MongoDB, a five-stage ETL pipeline, the CI/CD and the Linux box underneath it. A year at BRG writing Stata pipelines that shipped to opposing counsel to re-run. Now building tooling for agent work: provenance, human-in-the-loop gates, autonomous fleets — the machinery for checking what a model produced. I structure the work for coding agents where that pays, and write the code myself where it doesn't.

SKILLS

Applied AI Systems	LLM Orchestration; Provenance & Auditability; Multi-Agent Autonomous Loops; Human-in-the-Loop Gating; LLM Application Architecture
Languages	Python; JavaScript; Bash; R
Working Knowledge	SQL; Rust; Go; TypeScript; Lua; React
Systems & Backend	Flask; MongoDB; REST APIs; Click; Concurrency & DAG/Workflow Design; Linux Server Management
Infrastructure & DevOps	Linux; Docker; NGINX; GitHub Actions; CI/CD; Git; SSH; Playwright

SELECTED PROJECTS

Arachne

Apr 2026 – Present

- Designed an append-only provenance system in which every AI output references the source passages supporting each claim
- Directed a 13-crate Rust backend around a parallel DAG executor with lock-free single-owner scheduling and lossless pause/resume at human-in-the-loop gates, with a provider-agnostic LLM layer isolated in a sidecar process
- Specified a custom React node-graph canvas rather than adopting a node-editor library after react-flow proved a poor fit: the agent assembles a flow as an inspectable graph and a person verifies it before it runs
- Ran development as a multi-agent autonomous loop, unattended in Docker: task graph with preconditions and definitions-of-done, one git worktree per agent, hard egress allowlist. The fleet produced the implementation across 340+ tracked tasks

Switchboard

Mar 2026 – May 2026

- Directed a Go daemon that correlates each AI-agent process across three namespaces with no shared ID (/proc, the terminal multiplexer, and the Wayland compositor), anchored on the kernel-owned controlling TTY
- Approved its pidfd-based death detection (poll() on each PID's pidfd rather than scanning /proc or trapping signals) so a session is observed dead the instant it becomes a zombie
- Directed an agent-built Go rewrite of a ~25-file Bash/Lua prototype I had hand-written, once the hand-rolled single-writer daemon showed the problem needed real concurrency primitives

EXPERIENCE

Institute for Research on Labor and Employment

Jul 2024 – Dec 2025

Software & Data Engineer | Berkeley, CA

- Sole engineer on sspi.world, a full-stack research platform (Flask, MongoDB, JavaScript): built and ran it end to end, replacing a manual Google Sheets process with reproducible automated pipelines across 60+ sources, so coverage decided which countries the index could include, not the reverse
- Designed a five-stage DAG pipeline (collect, clean, compute, score, finalize) with fail-fast validation over several million records
- Built a Click CLI wrapping the app's HTTP endpoints for development and data exploration; in 2025 it became the structured interface my AI coding agents used to work inside the codebase
- Operated the platform the team depended on: shipped semver-tagged releases via GitHub Actions over SSH (atomic symlink swaps), root-caused MongoDB OOM kills, migrated Apache to NGINX, and led three research teams (~6 undergraduates each) over my years there

Berkeley Research Group

Aug 2023 – Jul 2024

Economics & Damages Associate | Emeryville, CA

- Wrote reproducible Stata analysis pipelines whose do-files reproduce every figure they report end-to-end from raw inputs, shipped as litigation backups subject to adversarial re-running by opposing experts; coverage/sanity/outlier checks caught real defects
- Built BinderBuilder, a Python CLI that cut footnote-checking on expert reports from ~60s to ~25s in the common case: TF-IDF parsing of the 200-500 footnotes in a document, fuzzy matching to the cited files, and auto-highlighting the passage so the reviewer verifies instead of searching

Institute for Research on Labor and Employment

Aug 2021 – Aug 2023

Research Data Analyst | Berkeley, CA

- Designed data-collection workflows across 60+ sources, the manual process the platform above later replaced

EDUCATION

University of California, Berkeley

Aug 2018 – May 2022

B.A. Applied Mathematics | B.A. Economics, Honors

Relevant coursework: Programming, Data Structures & Operating Systems (CS 61A/B/C); Artificial Intelligence (CS 188); Probability & Statistics (STAT 134/135)